

Mathematics 2 (5EZB0) Lecture 12

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Different line integrals

Line integral

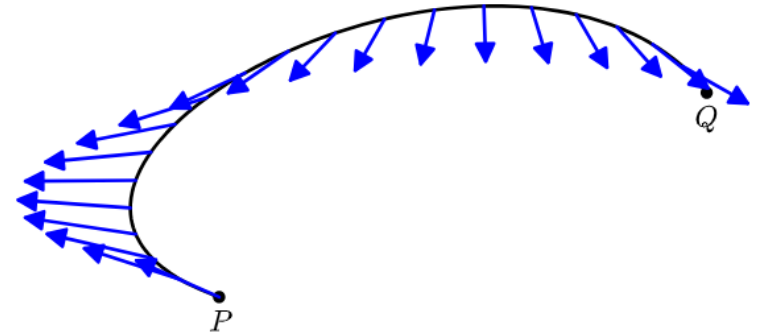
- Consider the parametric curve $\mathcal{C}$ (parameter $a \leq t \leq b$) in $\mathcal{D}$:
 $\mathbf{r}(t) = x(t) \hat{\mathbf{i}} + y(t) \hat{\mathbf{j}} + z(t) \hat{\mathbf{k}}$
- Consider the scalar and vector fields, $f: \mathcal{D} \rightarrow \mathbb{R}$, and $\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}^d$

Then,

$$\int_{\mathcal{C}} \mathbf{f}(\mathbf{r}) \cdot d\mathbf{r} = \int_a^b \mathbf{f}(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt} dt$$
$$\int_{\mathcal{C}} f(\mathbf{r}(t)) ds = \int_a^b f(\mathbf{r}(t)) \left| \frac{d\mathbf{r}}{dt} \right| dt$$

- If $\mathcal{C}$ is closed, then the line integral is called *the circulation*

$$\oint_{\mathcal{C}} \mathbf{f}(\mathbf{r}) \cdot d\mathbf{r}$$



Question 1

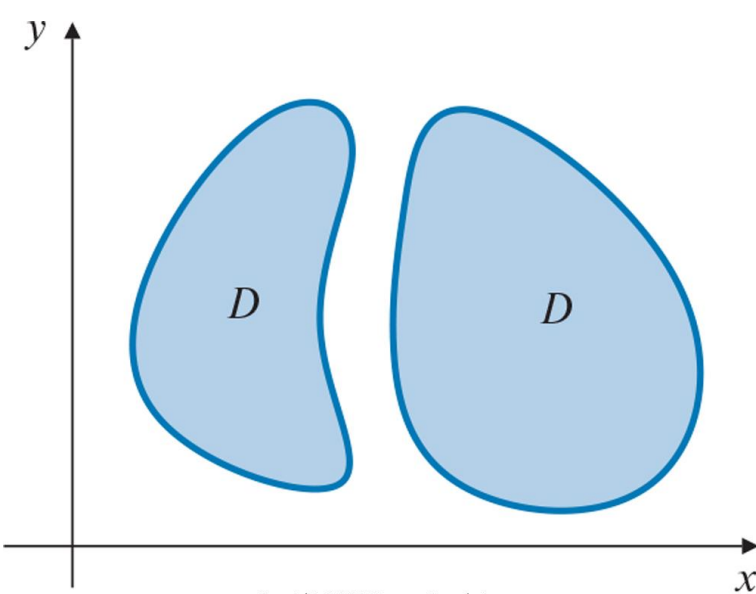
Consider the field

$$\mathbf{f}(x, y) = \frac{(-y \hat{\mathbf{i}} + x \hat{\mathbf{j}})}{x^2 + y^2}.$$

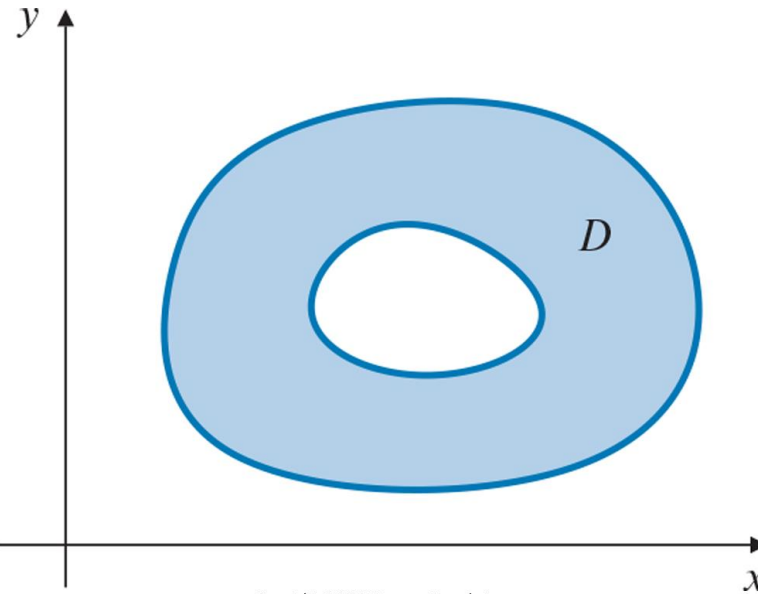
Find $\oint_{\mathcal{C}} \mathbf{f}(\mathbf{r}) \cdot d\mathbf{r}$ along the following curves $\mathcal{C}$:

1. The circle $x^2 + y^2 = 2$ counterclockwise from $(1, -1)$.
2. The square $[-1, 1] \times [-1, 1]$ counterclockwise from $(1, -1)$.

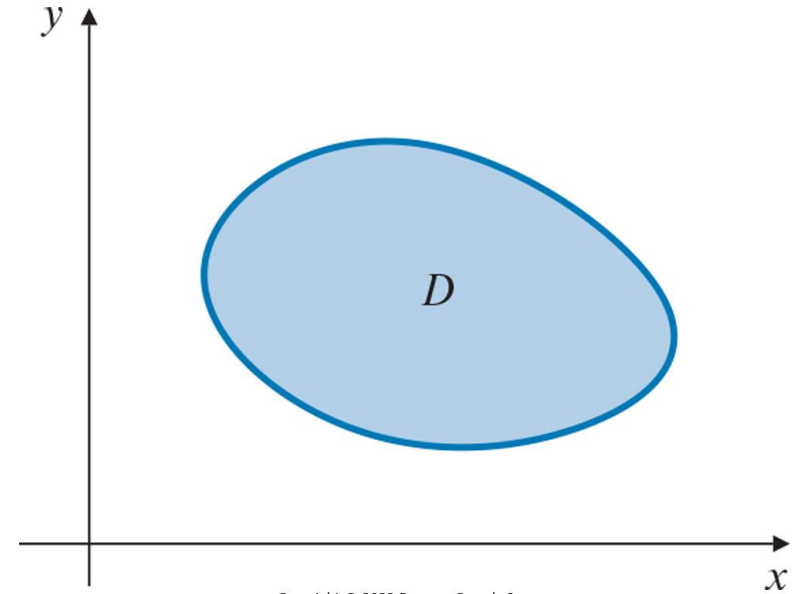
Conservative field



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disconnected domains

There exists two points which cannot be joined by a curve lying entirely in $\mathcal{D}$

Connected domains

A domain $\mathcal{D}$ is said to be connected if every pair of points $\mathbf{P}$ and $\mathbf{Q}$ can be joined by a piecewise smooth curve lying in $\mathcal{D}$.

Simply connected domains

A connected domain $\mathcal{D}$ in which every simple closed curve can be shrunk continuously to a point in $\mathcal{D}$ without ever passing out of $\mathcal{D}$.

Conservative field theorem

Conservative Field Theorem

Let $d \in \{2,3\}$, and $\mathcal{D} \subset \mathbb{R}^d$ be simply connected, and $\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}^d$ be a continuously differentiable field. Then the following statements are equivalent

- a) $\mathbf{f}$ is a conservative field, i.e., $\mathbf{f} = \nabla\varphi$.
- b) the integral $\int_{\mathcal{C}} \mathbf{f}(\mathbf{r}) \cdot d\mathbf{r}$ only depends on the endpoints of $\mathcal{C}$.
- c) The integral $\oint_{\mathcal{C}} \mathbf{f}(\mathbf{r}) \cdot d\mathbf{r} = 0$ for all closed $\mathcal{C}$
- d) for all $1 \leq i, j \leq d$,

$$\frac{\partial f_i}{\partial x_j} = \frac{\partial f_j}{\partial x_i}.$$

In other words, $D\mathbf{f}$ is symmetric, or $\nabla \times \mathbf{f} = \mathbf{0}$

Question 2

Show that $\mathbf{f}$ in Question 1 can be written as $\nabla\varphi$ and satisfies (d). Does it satisfy (c)? Justify your observations.

➤ Let $\mathbf{f} = \nabla\varphi: \mathcal{D} \rightarrow \mathbb{R}^d$ be a conservative field. Then, along a curve $\mathcal{C}$ connecting $\mathbf{r}(a) = \mathbf{P}$ and $\mathbf{r}(b) = \mathbf{Q}$ and entirely belonging in $\mathcal{D}$:

$$\int_{\mathcal{C}} \mathbf{f}(\mathbf{r}) \cdot d\mathbf{r} = \varphi(\mathbf{Q}) - \varphi(\mathbf{P}),$$

Parametric surfaces

Parametric surfaces

Let a point at a surface point be described by
 $\mathbf{r}(u, v) = x(u, v) \hat{\mathbf{i}} + y(u, v) \hat{\mathbf{j}} + z(u, v) \hat{\mathbf{k}}$

Examples

- The graph of a function $z = f(x, y)$ can be parametrically described by

$$\mathbf{r}(x, y) = x \hat{\mathbf{i}} + y \hat{\mathbf{j}} + f(x, y) \hat{\mathbf{k}}$$

- The ellipse $\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 = 1$:

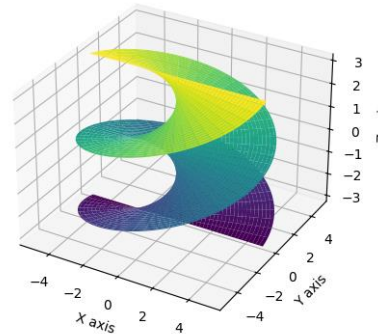
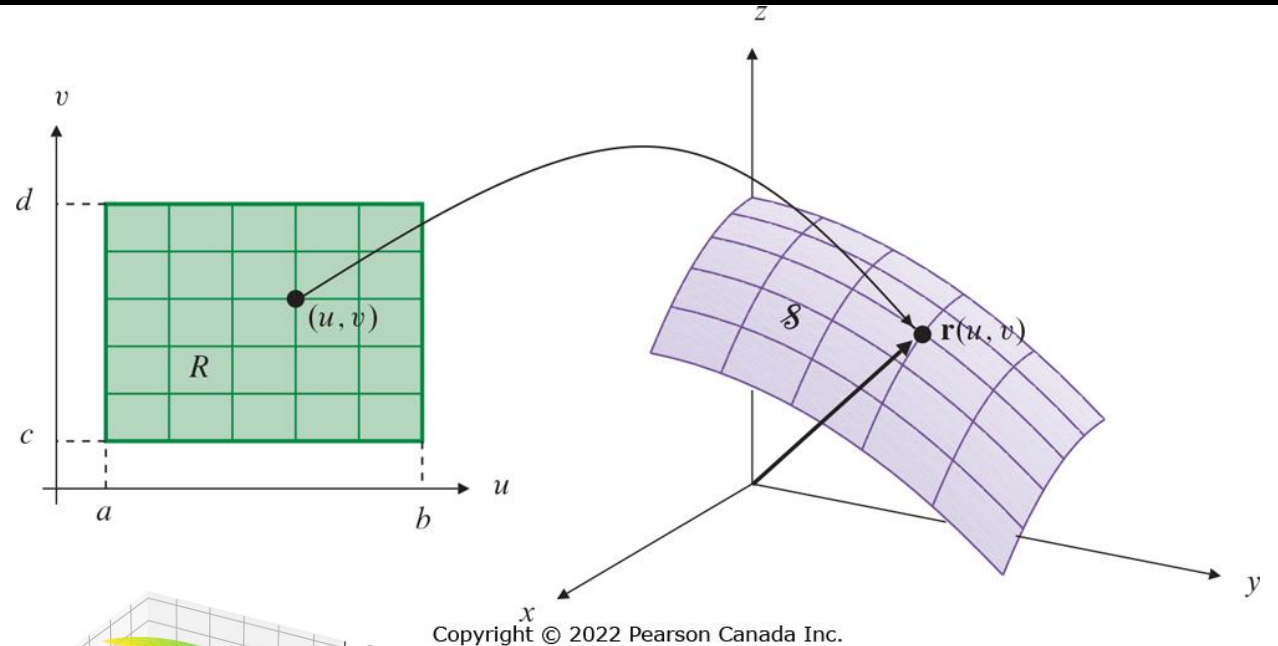
$$\mathbf{r}(u, v) = a \cos u \sin v \hat{\mathbf{i}} + b \sin u \sin v \hat{\mathbf{j}} + c \cos v \hat{\mathbf{k}}$$

for $0 \leq u \leq 2\pi, 0 \leq v \leq \pi$.

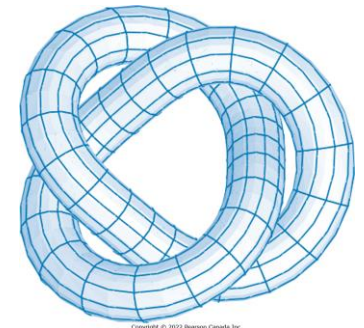
- One half of hyperboloid $\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = \left(\frac{z}{c}\right)^2 - 1$:

$$\mathbf{r}(u, v) = a v \cos u \hat{\mathbf{i}} + b v \sin u \hat{\mathbf{j}} + c \sqrt{1 + v^2} \hat{\mathbf{k}}$$

for $0 \leq u \leq 2\pi, v \geq 0$. Can you find some other parametrization of the above?



Run **Helicoid.py** to observe the surface of a helicoid
 $(u \cos v, u \sin v, v)$



More complicated surfaces can be described compared to only graphs

Surface integral

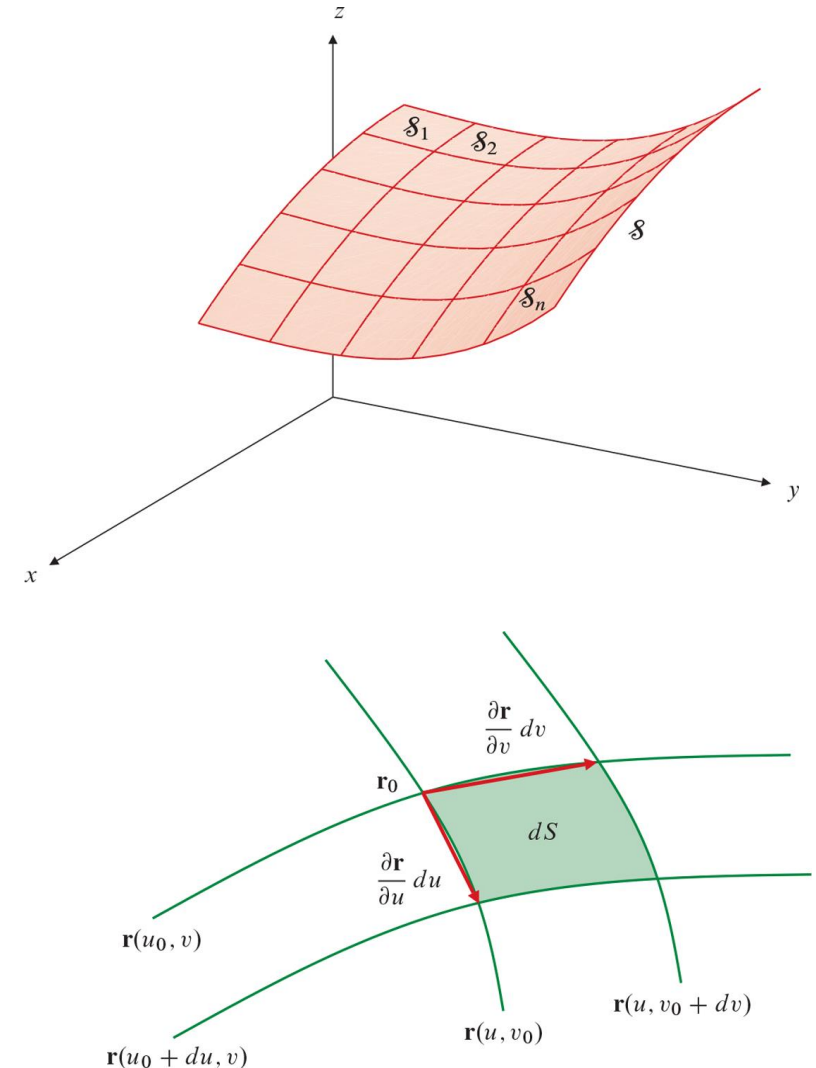
- Let $\mathbf{r}(u, v) = x(u, v) \hat{\mathbf{i}} + y(u, v) \hat{\mathbf{j}} + z(u, v) \hat{\mathbf{k}}$ describe a parametric surface $\mathcal{S}$ for $(u, v) \in \mathcal{D}$. Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ be a continuous function.
- The surface integral is interpreted using the Riemann sums, as before
- The infinitesimal area element spanned by changing u to $u + du$, and v to $v + dv$ has area

$$dS = \left| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right| du dv$$

$$\text{where } \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} = \det \begin{pmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \end{pmatrix} = \frac{\partial(y, z)}{\partial(u, v)} \hat{\mathbf{i}} + \frac{\partial(z, x)}{\partial(u, v)} \hat{\mathbf{j}} + \frac{\partial(x, y)}{\partial(u, v)} \hat{\mathbf{k}}$$

- The surface integral of f is then

$$\iint_{\mathcal{S}} f(x, y, z) dS = \iint_{\mathcal{D}} f(\mathbf{r}(u, v)) \left| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right| du dv$$



Surface integral

➤ Let $\mathbf{r}(u, v) = x(u, v) \hat{\mathbf{i}} + y(u, v) \hat{\mathbf{j}} + z(u, v) \hat{\mathbf{k}}$ describe a parametric surface $\mathcal{S}$ for $(u, v) \in \mathcal{D}$. Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ be a continuous function.

➤ The surface integral of f is then

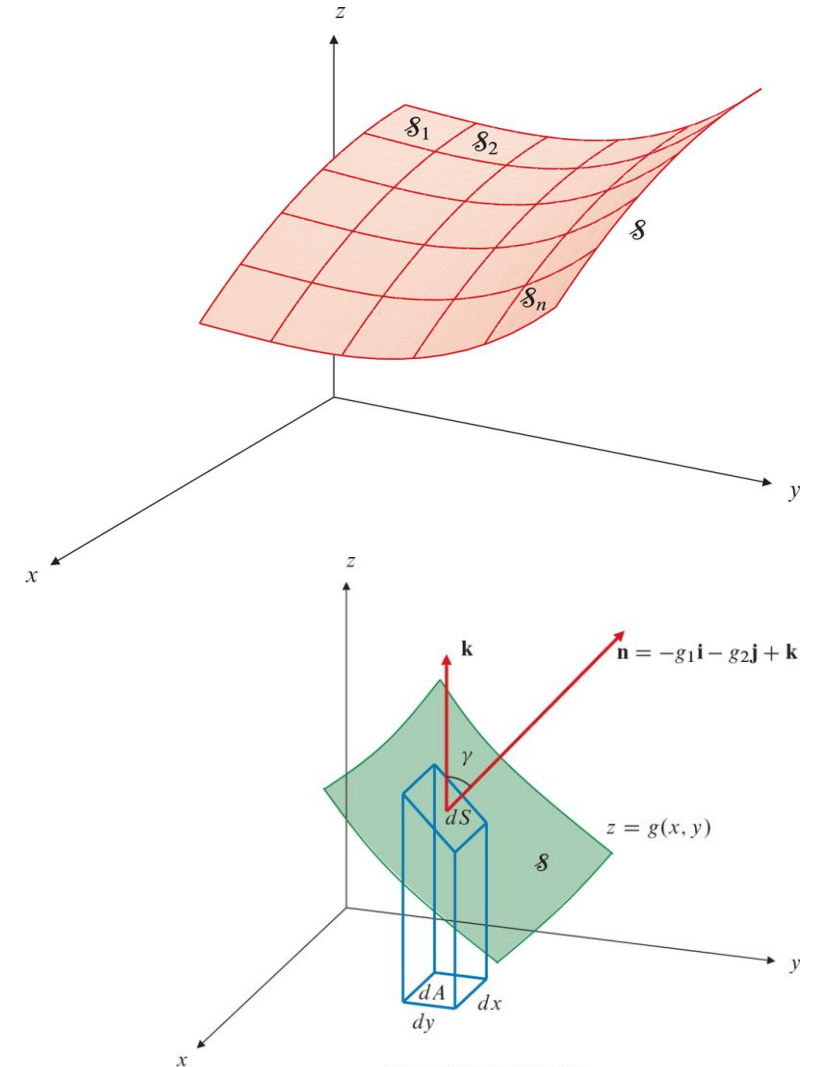
$$\iint_{\mathcal{S}} f(x, y, z) dS = \iint_{\mathcal{D}} f(\mathbf{r}(u, v)) \left| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right| du dv$$

where $\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} = \det \begin{pmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \end{pmatrix} = \frac{\partial(y, z)}{\partial(u, v)} \hat{\mathbf{i}} + \frac{\partial(z, x)}{\partial(u, v)} \hat{\mathbf{j}} + \frac{\partial(x, y)}{\partial(u, v)} \hat{\mathbf{k}}.$

Remark

- To find the area of $\mathcal{S}$, put $f = 1$
- The formula of area coincides with the formula derived before of surface area of a graph $\mathbf{r}(x, y) = x \hat{\mathbf{i}} + y \hat{\mathbf{j}} + f(x, y) \hat{\mathbf{k}}$,

$$|\mathcal{S}| = \iint_{\mathcal{D}} \sqrt{1 + |\nabla f(x, y)|^2} dx dy$$



Surface integral

- Let $\mathbf{r}(u, v) = x(u, v) \hat{\mathbf{i}} + y(u, v) \hat{\mathbf{j}} + z(u, v) \hat{\mathbf{k}}$ describe a parametric surface $\mathcal{S}$ for $(u, v) \in \mathcal{D}$. Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ be a continuous function.
- The surface integral of f is then

$$\iint_{\mathcal{S}} f(x, y, z) dS = \iint_{\mathcal{D}} f(\mathbf{r}(u, v)) \left| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right| du dv$$

where $\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} = \det \begin{pmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \end{pmatrix} = \frac{\partial(y, z)}{\partial(u, v)} \hat{\mathbf{i}} + \frac{\partial(z, x)}{\partial(u, v)} \hat{\mathbf{j}} + \frac{\partial(x, y)}{\partial(u, v)} \hat{\mathbf{k}}.$

Question 3.1

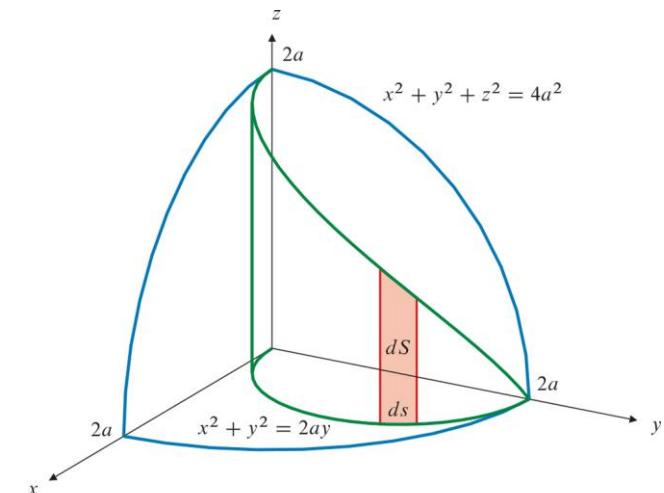
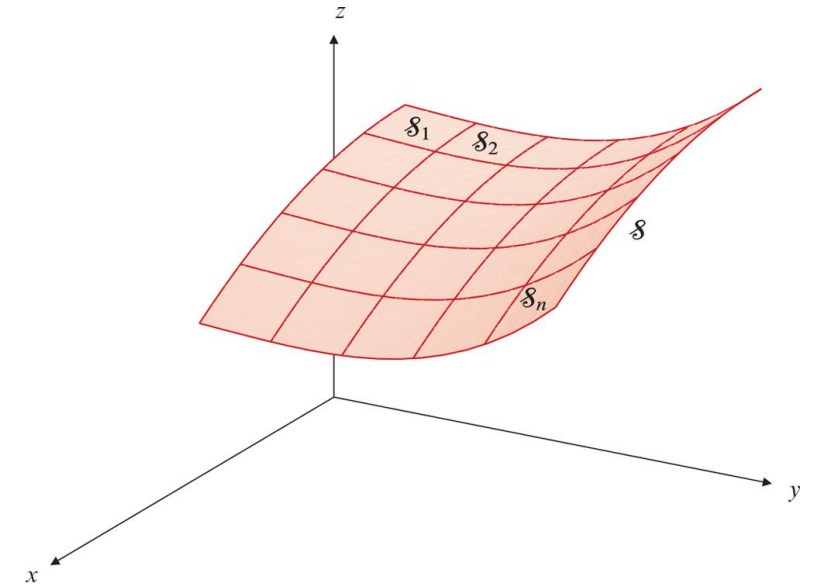
The moment of inertia of a uniform spherical shell:

$$\mathcal{S} = \{x^2 + y^2 + z^2 = a^2\}$$

$$\iint_{\mathcal{S}} (x^2 + y^2) dS$$

Question 3.2

Find the area of the part of the cylinder $x^2 + y^2 = 2ay$, that lies inside the sphere $x^2 + y^2 + z^2 = 4a^2$



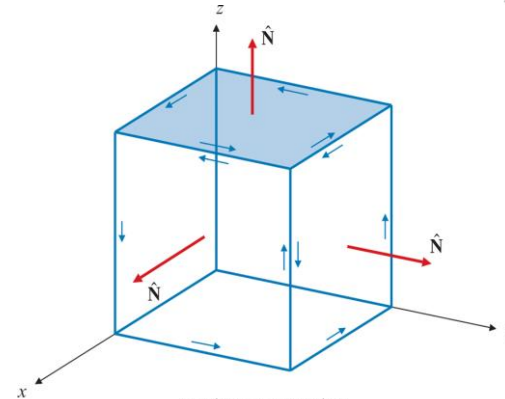
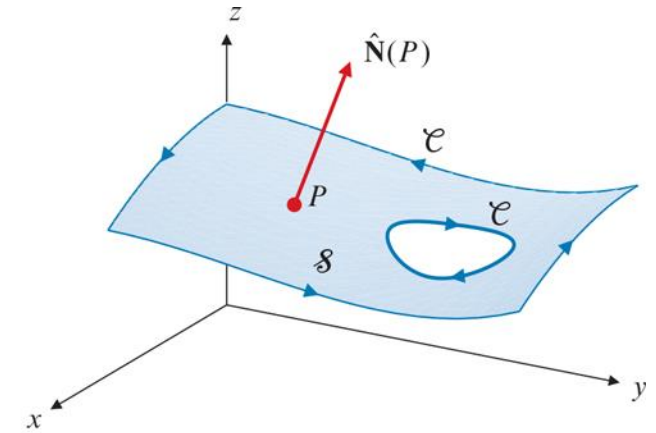
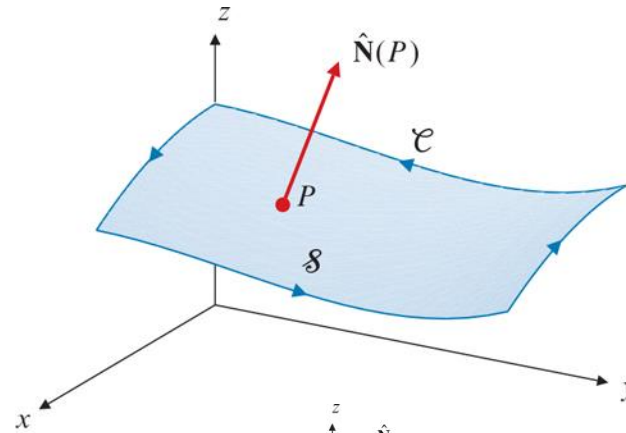
Oriented surfaces

Oriented surfaces

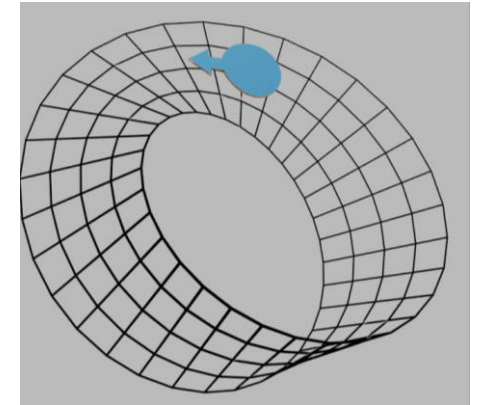
A smooth surface $\mathcal{S}$ is *orientable* if there exists a unit normal field $\hat{\mathbf{N}}: \mathcal{S} \rightarrow \mathbb{R}^3$ such that for any $\mathbf{P} \in \mathcal{S}$, $\hat{\mathbf{N}}(\mathbf{P})$ varies continuously with $\mathbf{P}$. The field $\hat{\mathbf{N}}$ is called an *orientation*.

The side which points along $\hat{\mathbf{N}}(\mathbf{P})$ is called the *positive side*, the other side is *negative side*.

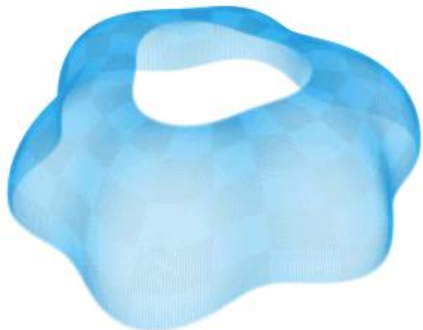
- An oriented surface $\mathcal{S}$ induces an orientation on any of its boundary curves $\mathcal{C}$ (clockwise and counter-clockwise)
- piecewise smooth surface is **orientable** if, whenever two smooth component surfaces join along a common boundary curve $\mathcal{C}$, they induce opposite orientations



A piecewise smooth orientable surface: *Cube*



A non-orientable surface:
The Möbius strip



Oriented surfaces

Oriented surfaces

A smooth surface $\mathcal{S}$ is *orientable* if there exists a unit normal field $\hat{\mathbf{N}}: \mathcal{S} \rightarrow \mathbb{R}^d$ such that for any $\mathbf{P} \in \mathcal{S}$, $\hat{\mathbf{N}}(\mathbf{P})$ varies continuously with $\mathbf{P}$. The field $\hat{\mathbf{N}}$ is called an *orientation*.

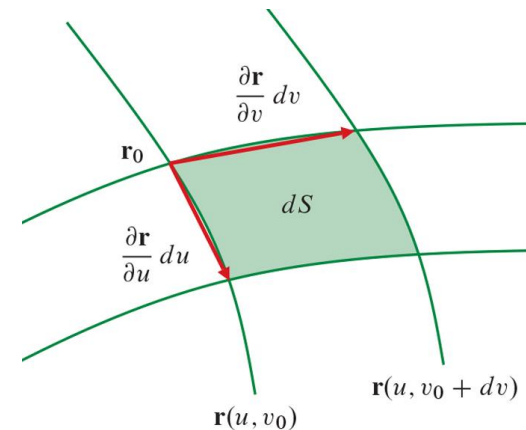
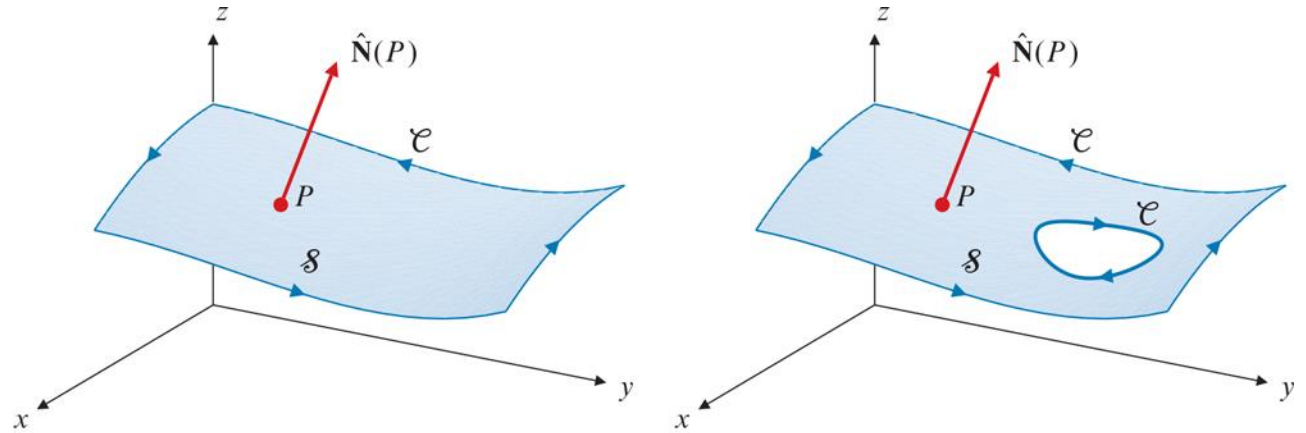
The side which points along $\hat{\mathbf{N}}(\mathbf{P})$ is called the *positive side*, the other side is *negative side*.

➤ For a parametric surface $\mathbf{r}(u, v)$:

$$\hat{\mathbf{N}}(\mathbf{r}(u, v)) = \frac{\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v}}{\left| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right|}$$

➤ For a graph $z = f(x, y)$

$$\hat{\mathbf{N}}(\mathbf{r}(u, v)) = \frac{-\nabla f + \hat{\mathbf{k}}}{\sqrt{1 + |\nabla f|^2}}$$



Flux integrals

- Let $\mathbf{r}(u, v) = x(u, v) \hat{\mathbf{i}} + y(u, v) \hat{\mathbf{j}} + z(u, v) \hat{\mathbf{k}}$ describe an orientable surface for $(u, v) \in \mathcal{D}$.
- Let $\hat{\mathbf{N}}: \mathcal{S} \rightarrow \mathbb{R}^3$ be the orientation
- Let $\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}^3$ be a vector field
- The vector surface area element is defined as

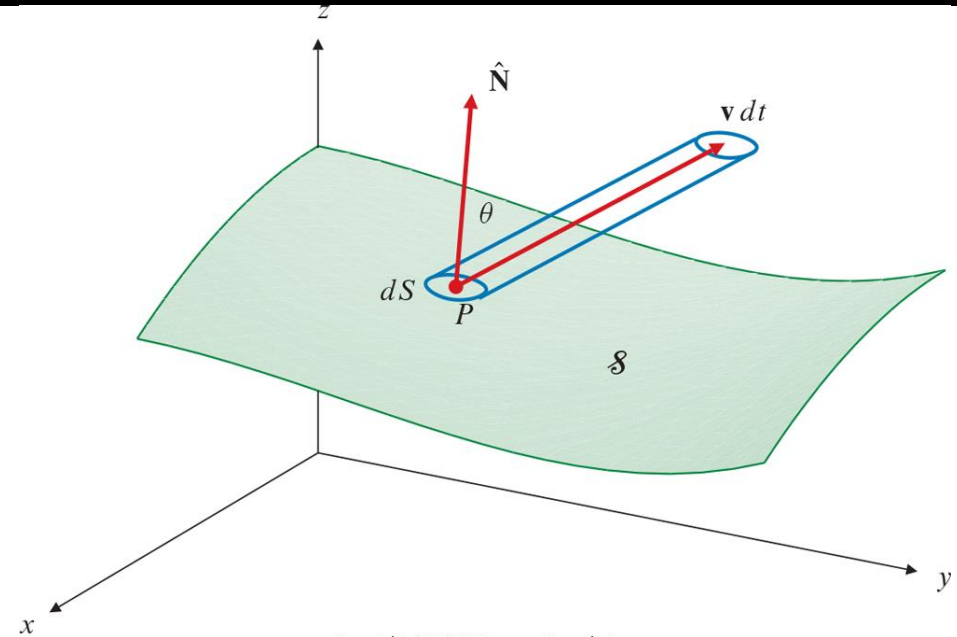
$$d\mathbf{S} = \hat{\mathbf{N}} dS = \left(\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right) du dv$$

- Then, the flux integral is

$$\iint_{\mathcal{S}} \mathbf{f} \cdot d\mathbf{S} = \iint_{\mathcal{D}} \mathbf{f}(\mathbf{r}(u, v)) \cdot \left(\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right) du dv$$

- If $\mathcal{S}$ is a closed surface, then the integral is written as

$$\oiint_{\mathcal{S}} \mathbf{f} \cdot d\mathbf{S}$$



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Question 4

For the gravitational field

$$\mathbf{F} = -\frac{C\mathbf{r}}{|\mathbf{r}|^3}$$

find the flux integral over the unit sphere.